

Class Incremental Website Fingerprinting Attack Based on Dynamic Expansion Architecture

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Abstract—Encrypted traffic on anonymizing networks is still at risk of being exposed to the Website Fingerprinting (WF) attack. This attack can seriously threaten the online privacy of users of anonymity networks such as Tor. While deep-learning-based WF attacks achieve high accuracy in controlled experimental settings, they cannot continuously learn after deployment. In real-world environments, new websites are constantly emerging, requiring attackers to expand their monitoring scope continuously. This necessitates attack models capable of continuous learning and expanding classification capabilities. In this paper, we explore how attackers can leverage incremental class learning techniques to continuously learn new classes while retaining the ability to distinguish old ones. This approach mitigates the catastrophic forgetting problem in dynamic, open-world scenarios. We introduce a new WF attack, Class Incremental Fingerprinting (CIF), which employs a scalable architecture enabling Class Incremental Learning (CIL) with limited resources. We evaluate this attack in various scenarios, such as learning 100, 200, and 500 monitored website classes across 5 and 10 incremental tasks, achieving an average accuracy of 97.8% and above. Additionally, we assess the CIF attack's effectiveness in open-world multi-classification scenarios and test it in few-shot settings using the proposed data augmentation method, Mixtam, achieving an average task accuracy of 87.6% and above with only 30 samples per class.

Index Terms—Tor, privacy, website fingerprinting, deep learning, class incremental learning.

I. INTRODUCTION

WITH the rapid development of the Internet, the privacy protection of users has become increasingly important. Over the past few decades, various anonymous communication systems have emerged, among which Onion Routing (Tor) [1] is the most popular. According to Tor's metrics [2], three to five million active users utilize Tor to protect their privacy every day. The Tor network employs multi-layer encryption and relay switching to protect user privacy. User data passes through multiple relay nodes in the Tor network, each of which can only decrypt a single layer of information, making user identity and activities difficult to trace. However, the Website Fingerprinting (WF) attack [3], [4], [5], [6], [7] is a

technique that identifies a specific website a user is visiting by monitoring the user's Internet traffic. Even if the traffic is encrypted, an attacker can infer which website a user is visiting by the patterns and characteristics of the traffic.

WF attacks are essentially multi-classification tasks. Attackers collect traffic traces for each website, analyze features, and train classifiers. Traditional methods for WF attacks rely on manual feature extraction [8], [9], [10], [11], requiring researchers to identify critical Web traffic features for training classifiers. These manual methods are time-consuming and do not cover all possible features, limiting their usability in practical applications. In contrast, deep learning methods [12], [13], [14], [15] can automatically learn feature representations from data without manual feature selection and adjustment, providing greater flexibility and improved generalization.

Deep learning-based WF attacks have achieved high performance in classical experimental scenarios. However, they still face challenges in practical applications, and their real-world effectiveness has been questioned by some criticisms [16], [17], [18]. Enhancing the real-world applicability of WF attacks and reducing reliance on idealized conditions have emerged as critical focal points for research. Multi-tab WF attacks [16], [19], [20] aim to overcome the single-page assumption by addressing scenarios where users open multiple tabs within a short period, generating overlapping traffic traces. Few-shot WF attacks [21], [22], [23] address the dependency on large training datasets. Some related researches [24], [25] focus on enhancing the robustness of the model to real-world challenges.

According to the Tor Metric [26], the official metric tool of the Tor Project, approximately 30,000 hidden services are published to the Tor network daily. As the number of websites a victim may visit increases, attackers must frequently expand their monitored websites, requiring WF attacks to demonstrate continuous learning capabilities. However, current research has not addressed how to enhance WF attack models' continuous learning and adaptation in real-world open-world settings, especially when adapting to new websites under limited computational and storage resources. Conventional deep learning-based WF attacks [12], [13], [15] use pre-collected data from a large set of target websites to train a complex model capable of recognizing all sample classes simultaneously. However, such a model can only generalize to known classes.

Updating the model with data from new classes results in catastrophic forgetting [27], which erodes the discriminative power of previously learned classes. Re-training the model

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from scratch incurs significant time and storage costs. Class incremental learning (CIL) in deep learning is uniquely suited to processing continuous information streams in real-world scenarios. It effectively absorbs new knowledge while adeptly retaining and optimizing previously acquired knowledge. With CIL, the attack model can acquire the ability to categorize traffic from new websites without compromising its discriminative power over existing website classes.

In this paper, we based on a dynamically scaling architecture [28], [29], [30], [31] propose a new WF attack method, Class Incremental Fingerprinting (CIF). Our methodology enables attackers to enhance their attack model's generalization capacity progressively. By retraining on new class traffic samples without retaining complete datasets of previous classes, this process prevents catastrophic forgetting of previously learned knowledge. Consequently, the model can adapt effectively to real-world, open, and evolving network environments, even under limited computational and storage resources.

Previous WF studies have not considered dynamically expanding site set scenarios. These methods are generally evaluated in closed-world and open-world settings, neither of which can simulate the dynamic, open-world environments of real-world scenarios or evaluate the incremental site fingerprinting attack methods we propose. In response, we introduce two novel scenario settings: dynamic closed-world and dynamic open-world.

The main contributions of our work are:

- We investigate the problem of WF attacks lacking continuous expansion learning capabilities in real open environments. To address this issue, we introduced a class incremental learning method to enhance the attack model's practical application ability. This approach enables attackers to continuously expand their monitoring scope across target websites without triggering catastrophic forgetting, while simultaneously not requiring a significant increase in computational or memory resource consumption.
- We propose two new and more challenging scenarios for evaluating CIF attacks: Dynamic closed-world and Dynamic open-world. In these scenarios, the attacker can continuously expand the set of monitored sites to update the model's identification and classification capabilities.
- We propose a new attack, Class Incremental Fingerprinting (CIF), for more realistic scenario applications. Our attack employs the Dynamic Extension Architecture (DEA) for CIL, which achieves an average of 97.8% and above accuracy in dynamic closed-world incremental learning tasks and demonstrates effectiveness in open-world environments.
- We evaluated the performance of the CIF attack through a series of experiments. The experimental results show that the CIF attack not only remains effective in dynamic open-world scenarios but also effectively addresses the challenge of few-shot incremental learning by combining with our proposed Mixtam data augmentation method.

The rest of this paper is organized as follows: we introduce the threat model in Section II and provide background and

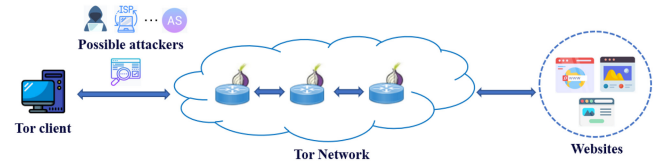


Fig. 1. The threat model of WF attacks.

as well as information about related work in Section III. We describe the proposed method in detail in Section IV and evaluate the proposed method experimentally in multiple scenarios in Section V. In Section VI, we discuss the limitations of the study as well as future directions. Finally, the work is summarized in Section VII.

II. THREAT MODEL

Tor is a widely used anonymous communication network. Users access the Onion Network through the free Onion Browser to browse websites while protecting their personal information. However, attackers can still compromise users' privacy by collecting traffic from monitored websites and utilizing Website Fingerprinting (WF) attack techniques to identify the websites users visit. As depicted in Fig. 1, akin to earlier WF attacks [8], [13], [32], within this threat model, the attacker is regarded as a localized passive observer. Positioned between the Tor client and the Tor ingress node (the access point to the Tor network), this attacker is capable of monitoring and logging network traffic without making any modifications or injections. The attacker's goal is to identify the specific websites visited by Tor users. Although traffic is encrypted through multiple layers of the Tor network, a WF attacker can still infer user behavior by analyzing features like timestamps and packet sizes using machine learning.

Closed-world and open-world scenarios are frequently employed to evaluate WF attack methods [12], [32], [33]. The closed-world scenario restricts the user's access to a subset of websites defined by the attacker, constituting the monitored sites. This scenario significantly differs from real-world conditions, but it provides a simplified and effective environment for analyzing and evaluating WF attack capabilities. The open-world scenario is more realistic, assuming a user can access both the monitored sites and those outside the attacker's scope, which form a substantial set of non-monitored sites. In the open-world multi-classification scenario, the attacker must determine if a traffic trace is part of the monitored set. If so, the attacker must pinpoint the specific site within the monitored set that the trace matches.

Neither scenario allows dynamic scaling of monitored websites to simulate a real open environment. Therefore, we propose two scenarios, dynamic closed-world, and open-world, to evaluate evolving WF attack scenarios.

III. BACKGROUND AND RELATED WORK

In the Tor network, messages are transmitted in fixed-size units (512 bytes) and are padded with zeros if insufficient data is available [15], [16]. An attacker can analyze Tor's traffic patterns and extract features such as interval time (IPD), direction, and packet size. These extracted features

can then serve as inputs for training classifiers to identify the websites visited by users. Broadly, WF attack methods discussed in the literature are categorized into two groups based on feature extraction techniques: traditional WF attacks that involve manual feature extraction and WF attacks that utilize deep learning for automated feature extraction.

Traditional website fingerprinting attack methods.

The earliest related research was conducted by Herrmann et al. (2009) [34], achieving an accuracy of merely 2.96%. Wang et al. [11] proposed a WF attack based on a manually extracted feature set, applying a k -NN classifier with feature weighting to emphasize the most important features. Pachinko et al. [33] introduced the CUMUL method, which accumulates Tor unit-sampling features and uses a Support Vector Machine (SVM) classifier to achieve a higher accuracy (91.38%) in a closed-world setting. Moreover, Hayes and Danezis [32] created the k -fingerprinting (k -FP) technique. This technique integrates a random decision forest and a k -nearest neighbor k -NN classifier to make use of a substantial collection of statistical features, showing high accuracy. While these traditional approaches based on machine learning techniques, such as SVM, (k -NN), and Random Forest, have achieved notable success, their reliance on hand-designed features and the lack of effective interaction between feature representation and classifier models have constrained their overall performance.

Deep website fingerprinting attack methods. Researchers have explored applying deep learning to WF attacks, inspired by its success in computer vision, natural language processing, and other fields. Deep learning methods enable direct end-to-end feature learning from raw data, removing the need for manual feature engineering. Rimmer et al. [12] employed only directional sequences from traffic traces as inputs and utilized three types of deep networks (SDAE, CNN, and LSTM) to achieve 96% classification accuracy in a closed-world scenario. Sirinam et al. [13] put forward the Deep Fingerprinting (DF) attack approach relying on a more intricate CNN structure. DF also used only directional sequences as inputs, achieving 98% accuracy in closed-world scenarios and effectively bypassing WTF-PAD defenses [35]. Bhat et al. [36] developed the Var-CNN model using Dilated Causal Convolutions on ResNets, demonstrating superior performance with few-shot training samples compared to DF. Deng et al. [37] proposed Holmes, a model using temporal and spatial distribution analysis to enhance WF attacks on partial traffic, achieving notable improvements in early-stage traffic recognition under dynamic network conditions. Attarian et al. [38] proposed a method that focuses on the first part of the tracked packets, achieving high preprocessing and classification efficiency while highlighting the critical role of directional information in website detection.

Few-shot website fingerprinting attack methods. Many researchers have proposed few-shot learning methods to overcome the limitation of requiring large labeled training datasets and enhance the effectiveness of WF attacks in real-world scenarios. Sirinam proposed the Triplet Fingerprinting (TF) [21] attack, which first trains a CNN model for feature extraction using a large dataset. The extracted features are

then used to train a k -NN classifier for WF classification. Experimental results demonstrate that the method performs well with limited data and is transferable by re-training the k -NN classifier with a new dataset. Zhou et al. [23] proposed the Clustered WF Attack (CWFA) method, which utilizes the cluster assumption to efficiently perform the attack in the presence of limited training data. NetCLR [39] employs self-supervised contrastive learning to augment Tor network traces, achieving robust WF accuracy and high transferability without relying on extensive labeled data.

Data augmentation techniques can also reduce reliance on large labeled datasets. Chen et al. [22] put forward a harmonized data augmentation (HDA) method. This method utilizes both within-sample and between-sample transformations to expand small training datasets. Additionally, Chen et al. [40] introduced the TForm-RF algorithm, which generates augmented traffic traces by extracting and randomizing traffic patterns from cell sequences, outperforming HDA in effectiveness.

Several studies have explicitly addressed the challenge of mixing multi-Web page traffic in WF attacks. These works [6], [19], [20], [41], [42] propose methods for segmenting and identifying traffic that combines tags from multiple Web pages. Additionally, Cherubin et al. [18] introduced a novel threat model that evaluates the practical effectiveness of WF attacks on the Tor network. This model leverages real Tor exit node traffic data and employs the TF approach to simulate more realistic attack scenarios.

Website fingerprinting Defenses. In addressing the challenge of WF attacks, the research community has explored and developed a variety of defense methods [3], [35], [43], [44], [45], [46], [47].

One defense strategy actively modifies the temporal and quantitative characteristics of communication traffic to complicate the correlation with specific website activities. Early attempts, such as BuFLO [48], CS-BuFLO [46], and Tamaraw [3], were effective in creating barriers to WF attacks but incurred significant bandwidth and time overheads due to fixed-rate packet delivery, adversely affecting user experience.

Subsequent research has sought to balance overhead reduction with defense effectiveness. For example, WTF-PAD [35] utilizes an adaptive approach to insert virtual packets during natural gaps in the actual data transmission, achieving a defense with zero time overhead. Front [49] also employs a zero-latency design that utilizes timestamps generated by the Rayleigh distribution to guide the embedding of the virtual packets. Walkie-Talkie [50] innovatively employs a half-duplex communication model, where the target-site data packets are mixed with non-target-site data, increasing the difficulty of identification for attackers. RegulaTor [51] minimizes time and bandwidth overheads through a well-designed packet-peaking adjustment strategy while maintaining defense efficiency. TrafficSliver [52] offers another approach by reconfiguring network architecture or employing client-side multiplexing, which reduces traffic exposure at single observation points without directly consuming bandwidth or time resources. More recent defenses, such as Surakav [53] and Palette [54], enhance protection by leveraging dynamic traffic

patterns and clustering techniques. Surakav employs GANs to generate adaptive, realistic traffic patterns, while Palette clusters similar traffic patterns into uniform groups, obfuscating traffic within each cluster and reducing identification risks.

In addition to these traffic obfuscation approaches, adversarial perturbations present another class of defense mechanisms. By introducing carefully crafted noise into the data, these methods aim to mislead deep learning models and hinder accurate classification. Typically, they [55], [56], [57], [58], [59] employ specialized loss functions to reshape distances in the feature space, offering alternative insights into the design of effective defenses. However, such approaches often require full trace data to compute the necessary perturbations and have shown limited effectiveness against adversarially trained models [60].

Class incremental learning. Class incremental learning (CIL) is a crucial branch of artificial intelligence that addresses the challenges of learning new classes. This approach ensures continuous knowledge accumulation without forgetting previous information. In recent years, incremental learning has achieved remarkable results in diverse fields, including computer vision [61], [62], [63], natural language processing [64], [65], and robotics [66], [67]. With rapid development, numerous approaches have emerged and can be classified into three categories: data-centric, model-centric, and algorithm-centric.

Data-centric approaches for mitigating forgetting rely heavily on additional data. For instance, GDumb [68] enables data replay to retain previous knowledge, while EWC [69] applies regularization to restrict parameter updates and ensure retention of learned tasks. FOSTER [70] uses model extension strategies, dynamically adding new modules to slow catastrophic forgetting, while integrating methods to reduce redundancy. Parameter regularization techniques like LWF [71] maintain performance on old tasks while acquiring new knowledge through customized loss functions.

Algorithm-centric approaches focus on knowledge transfer and model-tuning techniques. LUCIR [72] uses old model predictions as soft labels for knowledge distillation, reducing forgetting. In contrast, DER++ [73] combines replay buffering and knowledge distillation to optimize the update process, significantly suppressing forgetting.

IV. CIF ATTACK METHOD

This section details the proposed CIF attack. We first introduce the WF CIL problem setup and provide an overview of the methodology. Next, we describe the dynamic extension architecture for CIL, including the feature extractor, attention module, and knowledge distillation mechanism. Finally, we explain the data enhancement technique used in the few-shot CIL scenario.

A. Problem Setup and Method Overview

In the WF scenario, the CIL process assumes T batches of traffic fingerprint data $\{D_1, \dots, D_T\}$. The label space of the model at step T is defined as $\bar{W}_T = \bigcup_{i=1}^T W_i$, where W_i represents all labels in the dataset D_i at step i . Additionally,

the label sets at different steps are mutually exclusive, i.e., $W_i \cap W_j = \emptyset$ for any $i \neq j$. The model learns all class knowledge over T steps. At step T , the incoming training dataset is $D_t = \{(x_1^t, y_1^t), \dots, (x_n^t, y_n^t)\}$, where x_i^t denotes the i -th input traffic trace, $y_i^t \in W_t$ is its corresponding label, and n is the number of training samples. The goal of the model is to accurately predict all classes in W . This requires learning new class knowledge from step T data D_t while retaining previously learned knowledge from $\{D_1, \dots, D_{t-1}\}$.

To achieve this, our approach adopts a rehearsal strategy [27], [74], which saves representative class samples as a memory M_t within a fixed-size budget for subsequent training tasks. An iterative method selects samples based on the mean of the class feature space, ensuring an even distribution across the dataset and retaining representative samples for each class. In contrast, the non-rehearsal setting prohibits using data from previous tasks.

B. Class Incremental Learning Architecture

We propose a model architecture tailored for incremental learning of WF attacks, inspired by previous works [28], [29], [30]. The dynamic expansion framework (CIF) incrementally adds feature extractors and classifiers to handle progressively increasing classification tasks. At each incremental learning stage, feature extractors generate feature maps, which are enhanced through a fusion module employing the SE (Squeeze and Excitation) mechanism to highlight critical features selectively. To address forgetting, CIF employs knowledge distillation to integrate newly learned and existing knowledge. Furthermore, CIF uses a re-scoring mechanism to dynamically adjust classifier outputs, ensuring balanced performance between new and old classes. Finally, the loss function combines cross-entropy loss, knowledge distillation loss, and re-scoring loss to ensure effective learning of new tasks while retaining previously acquired knowledge.

It is shown in Fig. 2. The architecture is roughly divided into two phases in each CIL step:

- *First phase.* The data for the current task is input into the feature extractor to generate the corresponding feature maps. During subsequent learning steps, these feature maps are fused with those produced by the old feature extractor. This fusion effectively integrates new and old task knowledge, ensuring a comprehensive representation that supports subsequent classification and recognition tasks.
- *Second phase.* Feature maps are processed by the SE attention model, which automatically learns feature channel weights to enhance informative features and suppress irrelevant ones. Following this, the classifier re-scoring mechanism refines the ordering and scores of the classifier outputs, improving the model's output quality and adaptability to diverse task requirements.

Basic Architecture. The initial feature extractor $\mathcal{F}_1$ and classifier $\mathcal{C}f_1$ are trained on CIF following standard incremental learning methods [28], [75]. During the first step, only the classification loss $\mathcal{L}_{clf}$, computed using cross-entropy, is used to train both the feature extractor and the classifier. Assuming a total of n CIL steps, at each incremental step $t \in \{2, \dots, N\}$, a

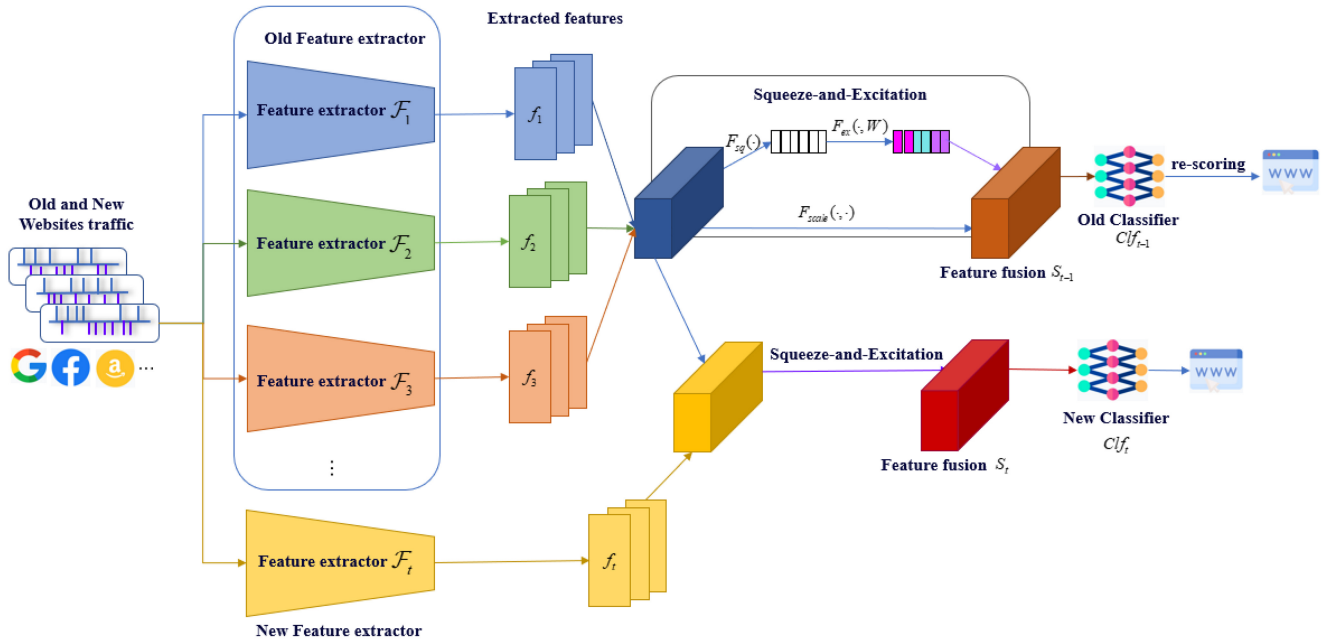


Fig. 2. An illustrative framework of the proposed CIF. It incrementally adds task-specific feature extractors and classifiers, enhances features using the SE mechanism, and balances new and old task performance via knowledge distillation and a re-scoring mechanism.

new feature extractor $\mathcal{F}_t$ and classifier $\mathcal{Clf}_t$ are introduced. The parameters of the existing feature extractors $\{\mathcal{F}_1, \dots, \mathcal{F}_{t-1}\}$ and the previous classifier $\mathcal{Clf}_{t-1}$ remain unchanged, while $\mathcal{Clf}_t$ is initialized using $\mathcal{Clf}_{t-1}$.

For each incremental step, the newly added feature extractor $\mathcal{F}_t$ learns knowledge from the current training data D_t and the fused features v_{t-1} derived at step $t-1$ via knowledge distillation. Specifically, the fusion feature v_t for traffic trace x (where $x \in \{D_1, \dots, D_t\}$) is obtained by concatenating feature vectors from all feature extractors. For a given x , features are extracted from all feature extractors and concatenated to produce v_t , as shown below:

$$v_t = [\mathcal{F}_1(x), \dots, \mathcal{F}_t(x)]. \quad (1)$$

For the fused features v_t , the feature fusion module S_t , based on the SE attention mechanism, enhances significant features to obtain a stronger feature representation f_t^{se} . This enhanced representation is passed into the new classifier $\mathcal{Clf}_t$, which produces the output logits $l_t(x)$. During this process, knowledge distillation guides $\mathcal{Clf}_t$ in preserving knowledge of the old classes.

A classifier re-scoring module is employed to avoid the possibility that the newly introduced classifier $\mathcal{Clf}_t$ may be biased against the old classes during inference (since it is primarily trained using samples from the new class). This module mitigates class imbalance by adjusting the weight distributions between old and new classes at the end of each training step.

Finally, the re-scored logits $l_t(x)$ are passed through a softmax function to produce the final classification result $\hat{y}$ for the traffic trace x .

$$\hat{y} = \arg \max(\text{Softmax}(l_t(x))), \quad (2)$$

the $\arg \max$ function is used to select the website class label with the highest probability as the prediction result $\hat{y}$.

Knowledge Distillation (KD). Knowledge distillation (KD) [29], [76] is an effective technique for supporting incremental learning tasks. In our approach, KD is applied to prevent catastrophic forgetting of previously acquired knowledge while accommodating new information, thus facilitating the accumulation and integration of knowledge across distinct tasks. KD consists of two components: feature-level and logit-level knowledge distillation. These components are applied to the new feature extractor $\mathcal{F}_t$ and classifier $\mathcal{Clf}_t$ at each incremental step to help them better differentiate between old and new classes and consolidate knowledge from previous tasks.

Feature-level knowledge distillation guides the training of the new feature extractor $\mathcal{F}_t$ by using the traffic trace and the old feature extractor $\mathcal{F}_{t-1}$ in the retained sample memory $\mathcal{M}_t$ as the teacher model. This strategy minimizes the difference between the output features of $\mathcal{F}_{t-1}$ and $\mathcal{F}_t$, enabling $\mathcal{F}_t$ to capture critical feature representations related to previous tasks, rather than relying solely on data from new tasks. We calculate the distance between the output features of $\mathcal{F}_{t-1}$ and $\mathcal{F}_t$ as the feature-level knowledge distillation loss:

$$\mathcal{L}_f(x) = \sqrt{\sum_{j=1}^d \left(F_t^{(j)}(x) - F_{t-1}^{(j)}(x) \right)^2}. \quad (3)$$

Logit-level knowledge distillation is performed by comparing, for each traffic trace, the logits outputs $l_t(x)$ and $l_{t-1}(x)$ from the classifiers $\mathcal{Clf}_t$ and $\mathcal{Clf}_{t-1}$, respectively. The Kullback-Leibler (KL) divergence [76] is employed to measure the discrepancy between the outputs of the two classifiers,

acting as the loss function to be minimized

$$\mathcal{L}_l(\mathbf{x}) = \sum_{w=1}^{\tilde{W}_{t-1}} p_w(\mathbf{x}) \log\left(\frac{p_w(\mathbf{x})}{q_w(\mathbf{x})}\right), \quad (4)$$

where, $q_w(x) = \frac{e^{\hat{l}_w(x)/T}}{\sum_{j=1}^{\tilde{w}_{t-1}} e^{\hat{l}_j(x)/T}}$, $p_w(x) = \frac{e^{l_w(x)/T}}{\sum_{j=1}^{\tilde{w}_{t-1}} e^{l_j(x)/T}}$, T is the temperature scalar, which controls the decision-making process of the model so that it balances between determinism and stochasticity. $\hat{l}_w(x)$ is a component of $l_{t-1}(x)$. The vector $l_{t-1}(x)$ defined as $l_{t-1}(x) = (\hat{l}_1(x), \dots, \hat{l}_{\tilde{w}_{t-1}}(x))^T$, represents the logits of the classifier Clf_{t-1} . $l_w(x)$ is an element of $l_t(x)$, $l_t(x) = (l_1(x), \dots, l_{\tilde{w}_{t-1}}(x), l_{\tilde{w}_{t-1}+1}(x), \dots, l_{\tilde{w}_t}(x))^T$ denotes the logits of the classifier Clf_t .

This process encourages Clf_t to closely replicate the probability distribution of the old class outputs generated by Clf_{t-1} . By achieving this, knowledge transfer from the old tasks is facilitated, and the retention of previously learned knowledge is preserved, effectively mitigating the effects of catastrophic forgetting. Consequently, a feature-level knowledge distillation loss $\mathcal{L}_f(x)$ and a logit-level knowledge distillation loss $\mathcal{L}_l(x)$ are introduced, forming the overall dynamically balanced knowledge distillation loss $\mathcal{L}_{d-kd}$:

$$\mathcal{L}_{d-kd} = w_1 \sum_{x \in \mathcal{R}} \mathcal{L}_f(x) \times \min\left(1, \frac{\text{epoch}}{w_{\text{warm up}}}\right) + w_2 \sum_{x \in \mathcal{R} \cup \mathcal{D}_t} \mathcal{L}_l(x), \quad (5)$$

where epoch denotes the current training epoch, and w_{warmup} defines the duration of the warm-up phase. The feature-based loss $\mathcal{L}_f(x)$ increases in importance as training progresses, with its contribution controlled by a dynamic weighting mechanism that adjusts its influence based on the training epoch. In contrast, the logit-based loss $\mathcal{L}_l(x)$ is applied consistently throughout the training process, ensuring stability in knowledge transfer across both current and past tasks. This combined loss function updates the parameters of the feature extractors and classifiers during training, facilitating information propagation between old and new tasks. It enables the model to better distinguish the differences between classes in both the old and new tasks.

SE Module (Squeeze-and-Excitation). Considering that the old feature extractor $\{\mathcal{F}_1, \dots, \mathcal{F}_{t-1}\}$, trained before the new learning step, cannot effectively extract features for traffic traces from the new step $\{t, \dots, T\}$, the quality of extracted features is affected. To address this issue, we adopt the SE attention module [77] to optimize the feature representation. Given the fused feature map $u_t \in \mathbb{R}^{C \times H \times W}$ as input, the SE attention mechanism first applies global average pooling (i.e., the Squeeze operation) to compress the spatial dimensions and obtain the compressed feature vector z :

$$z = \text{AvgPool}(u_t). \quad (6)$$

The compressed feature vector undergoes processing via a fully connected layer (MLP) to reduce its dimensionality. This is succeeded by dimensionality enhancement (namely the

Excitation operation), which is achieved by utilizing the ReLU activation function and another fully connected layer:

$$F_{ex}(z, W) = \sigma(W_2 \delta(W_1 z)), \quad (7)$$

where W_1 and W_2 are the weight matrices of the MLP, δ represents the ReLU activation function, and σ denotes the sigmoid activation function.

Finally, the output feature map is weighted by channel-wise multiplication through the sigmoid function and combined with the original input feature map using a residual structure. The overall feature fusion process of the SE attention module is described as:

$$f_{se}(u_t) = \sigma(W_2 \delta(W_1 z)) \otimes u_t + u_t, \quad (8)$$

where $\otimes$ represents element-wise multiplication.

The SE module enables the class incremental architecture to adaptively adjust the response of each channel, effectively enhancing the representation of useful features while suppressing irrelevant ones. This improves the feature expression and the ability to generalize the class incremental architecture.

Classifier Re-scoring (CR). In incremental learning, the model tends to focus excessively on novel classes, resulting in output logits biased towards these newer classes. This bias adversely impacts the model's generalizability and its ability to effectively transfer knowledge. To mitigate this issue, we employ the Classifier Re-scoring approach [78] to adjust the scores assigned to both old and new classes. This mechanism begins by computing the weight norms of the classifier for existing and newly introduced classes:

$$\begin{aligned} \text{norm}_{\text{old}} &= \left(\sqrt{\text{tr}(\theta_1^\top \theta_1)}, \dots, \sqrt{\text{tr}(\theta_{\mathcal{W}_{t-1}}^\top \theta_{\mathcal{W}_{t-1}})} \right), \\ \text{norm}_{\text{new}} &= \left(\sqrt{\text{tr}(\theta_{\mathcal{W}_{t-1}+1}^\top \theta_{\mathcal{W}_{t-1}+1})}, \dots, \right. \\ &\quad \left. \sqrt{\text{tr}(\theta_{\mathcal{W}_t}^\top \theta_{\mathcal{W}_t})} \right), \end{aligned} \quad (9)$$

Using the above norms, we compute the reweighting coefficient γ for classifier re-scoring:

$$\gamma = \frac{\text{Mean}(\text{norm}_{\text{old}})}{\text{Mean}(\text{norm}_{\text{new}})}, \quad (10)$$

where $\text{Mean}(\cdot)$ represents the mean value of the elements in the vector. By applying the reweighting factor γ , the logits are adjusted to align the mean norm of the novel classes with that of the old classes. The logits $l(x)$ of Clf_t are expressed as $l(x) = (l_{t-1}(\mathbf{x}), l_t(\mathbf{x})[\tilde{W}_{t-1} + 1, \dots, \tilde{W}_t])$. The rectified logits $l_{\text{re}}(x)$ are then given by:

$$l_{\text{re}}(x) = (l_{\text{old}}(x), \gamma \cdot l_{\text{new}}(x)). \quad (11)$$

Loss Function. The total training loss for WF CIL consists of three components: the classification loss $\mathcal{L}_{clf}$, computed using cross-entropy, the dynamically balanced knowledge distillation loss $\mathcal{L}_{d-kd}$, and the multi-class auxiliary loss $\mathcal{L}_{\mathcal{MH}}$. The auxiliary loss $\mathcal{L}_{\mathcal{MH}}$ is inspired by previous CIL work [28]

and is specifically designed to distinguish between old and new classes. It is defined as follows:

$$\mathcal{L}_{\mathcal{MH}} = -\frac{1}{|\mathcal{D}_t|} \sum_{x \in \mathcal{D}_t} \log \left(\frac{\exp(\mathbf{w}_{y'} \cdot \mathcal{F}_t(x))}{\sum_{k=1}^{|\mathcal{Y}_t|+1} \exp(\mathbf{w}_k \cdot \mathcal{F}_t(x))} \right). \quad (12)$$

This formula guides the feature extractor $\mathcal{F}_t(x)$ to learn discriminative features for both new and old classes. Here, $\mathcal{D}_t$ is the current training dataset, combining new data and memory data. The weight vector $\mathbf{w}_{y'}$ corresponds to the adjusted label y' , where new classes retain their original labels and old classes are mapped to a unified old class label. The Softmax normalization term in the denominator ensures the output is probabilistic. This loss term helps mitigate catastrophic forgetting by encouraging the feature extractor to preserve knowledge from previous tasks while adapting to new ones.

The total loss is formulated as:

$$\mathcal{L} = \mathcal{L}_{clf} + \mathcal{L}_{d-kd} + \mathcal{L}_{\mathcal{MH}}. \quad (13)$$

C. Data Representation

Our approach utilizes a Traffic Aggregation Matrix (TAM) [25], which differs from most prior deep learning-based WF attack methods that primarily focus on sequences of direction information in Tor traffic data. TAM aggregates multidimensional information, including direction, volume, and timing, providing a more comprehensive and robust set of features.

The network traffic trace related to a website visit can be expressed as $w = (s_1, s_2, \dots, s_l)$, where l denotes the length of the trace. We set $l = +\infty$ to account for all packets. Let $s_i = \langle t_i, d_i \rangle$ be a pair consisting of the packet timestamp and its direction. Here, t_i and d_i signify the arrival time and the direction of the i -th packet, respectively. It should be noted that $d_i \in \{\pm 1, 0\}$, with $+1$ denoting the outgoing packet direction and -1 representing the incoming packet direction.

The specific implementation of TAM involves partitioning the network traffic into multiple fixed-length time slots and counting the number of outgoing and incoming packets within each slot. Let M be the TAM of trace W , with N being the count of time slots taken into account in the TAM. Given that the length of each time slot is s , the maximum load time T encompassed by a trace can be computed as T/s . In the resulting two-dimensional matrix M , each row represents the packet counts for either the outgoing or incoming direction, while the columns correspond to different time windows. Each element m_{ij} records the cumulative count of packets in the respective direction within the time interval from $(j-1) \times s$ to $j \times s$ seconds.

TAM captures intermediate granularity information by aggregating outgoing and incoming packet counts per time slot, thereby enhancing robustness against various defense mechanisms. Furthermore, compared to using direction sequences alone as a traffic representation, TAM captures a richer set of traffic characteristics, improving the model's generalization capabilities. This aids in maintaining model performance during incremental learning processes.

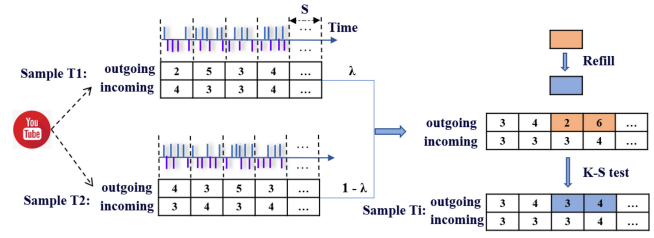


Fig. 3. Mixtam Data Augmentation Workflow for Few-shot CIL Scenarios.

D. Feature Extractor

We use RF [25] as a feature extractor in a dynamically scalable architecture for WF attacks. The RF model is specifically designed based on the CNN architecture, targeting the TAM traffic representation method. Leveraging the construction of TAM and a multi-level convolutional network structure, the model effectively extracts local features in both the temporal and spatial dimensions of the traffic flow. Furthermore, it refines robust features capable of resisting various defense strategies.

Convolutional Blocks. In the model, 2D convolutional blocks are initially applied to process the two-dimensional input data, namely the TAM. The TAM contains information about the number of data packets in different directions (upload, download) within time slots during network communication, reflecting the interaction between clients and servers and the variability of traffic patterns. Through two 2D convolutional layers, the model captures and extracts local and intermediate spatiotemporal features. These layers are combined with the ReLU activation function to enhance non-linear expressiveness, BatchNormalization layers to ensure stable convergence during training and prevent overfitting, and MaxPooling layers for downsampling. The TAM is then fused into a 1D feature map using a 2×2 maximum pooling layer, further reducing the feature dimensionality while preserving essential information. To extract higher-level features, the model reduces the number of channels and reshapes the output to expand the feature map size, followed by two 1D convolutional blocks for deeper feature mining.

Global Average Pooling (GAP). To avoid overfitting caused by the excessive number of parameters introduced by traditional fully connected layers, the model adopts the GAP [79] layer as an alternative. The GAP layer does not introduce additional parameters but directly computes the average of each feature map, reducing the model's complexity and enhancing its generalization ability. Finally, by flattening the vector obtained from averaging each feature map and passing it through the softmax function, the model generates a probability distribution for different website classes, completing the classification task.

E. Data Augmentation

In practical WF attack scenarios, obtaining sufficient website traffic samples for model training can be challenging, particularly in rapidly evolving network environments where timely collection of large volumes of data for model updates becomes an issue [21], [22], [40], [80]. In the few-shot

CIL scenario, we have devised a novel data augmentation technique, Mixtam, tailored for the TAM matrix-based representation of website traffic.

Mixtam employs a stochastic mixing [81], [82] enhancement mechanism based on random sampling from a Beta distribution. Specifically, this method selects two distinct samples from within the same category label of multiple traffic sequences. It then linearly interpolates these paired samples using a mixing coefficient λ drawn from a Beta distribution. This fusion process generates new, mixed traffic samples. As illustrated in Fig. 3, a linear interpolation method is applied to two samples $\vec{T}_{o1}$ and $\vec{T}_{o2}$ of the same website to compute the new virtual sample seed $\vec{T}_{oi}$:

$$\vec{T}_{oi} = \lambda \vec{T}_{o1} + (1 - \lambda) \vec{T}_{o2}, \quad (14)$$

where λ is chosen as in [81], using a random variable drawn from a Beta distribution $\beta(\alpha, \alpha)$, with α set to 0.1 after several tests.

If the relative change introduced by this mixing exceeds a predefined threshold, it indicates that data within that time window requires adjustment. Upon detecting deviations above this threshold, our approach probabilistically decides—based on the inherent characteristics of the raw data—whether to retain the augmented data values or revert to the original data values for filling. Let the original traffic TAM be represented by O , and the enhanced, synthetically generated TAM be designated as T . For each arbitrary time slot indexed by i , we determine the rates of change in the number of outgoing data packets and the volume of incoming traffic, yielding values Δ_{in} and Δ_{out} . These derived rates are then compared against a predetermined local variation threshold δ :

$$\Delta_{in}(i) = \frac{|T_{in}(i) - O_{in}(i)|}{|O_{in}(i)| + 1} \quad (15)$$

$$\Delta_{out}(i) = \frac{|T_{out}(i) - O_{out}(i)|}{|O_{out}(i)| + 1}. \quad (16)$$

The Kolmogorov-Smirnov (K-S) test [83], [84] is then utilized to assess whether the augmented data (T_{am}) statistically conforms to the distribution of the original data (O_{tam}). It computes the K-S statistic and its corresponding p-value for the two sample sets:

$$D = \sup_x |T_{or}(x) - T_{au}(x)|. \quad (17)$$

If the p-value falls below a predetermined significance level of 0.05, this suggests a significant difference between the distributions of the augmented and original datasets. In such cases, the augmented samples undergo further randomized adjustments to better align their distribution with the original data.

In few-shot CIL, we utilize the Mixtam algorithm to generate virtual trace samples and combine these virtual samples with the original samples to form a training set, thereby enhancing the model's robustness in situations where collecting a significant amount of training samples is not feasible.

V. EXPERIMENTAL EVALUATIONS

In this section, we conduct a series of experimental evaluations to investigate the performance of CIF attacks under

various experimental scenarios. Traditional deep learning methods [12], [13], [14], [15], [36], [38] must be retrained from scratch to satisfy the attacker's need to dynamically expand the set of monitoring sites. Moreover, no prior WF attack methods have considered similar scenarios for experimental evaluation. Recent WF attack methods [19], [22], [23], [24], [25], [40], [85], [86] also fail to address the lack of continuous learning capability in attack models, making them unsuitable for CIL scenarios.

In the absence of methods specifically designed for incremental learning in WF attacks, we conduct comparative experiments using TF [21], DF [13], Var-CNN [36], and NetCLR [39] under the same experimental setup. As a classic few-shot WF attack method, the TF attack exhibits certain transfer properties, enabling classification on an expanded set of websites by retraining the k -NN classifier. Similarly, NetCLR leverages fine-tuning to adapt to new website sets, demonstrating transferability across different data distributions.

The deep learning-based WF attack model Var-CNN [36] achieves superior performance compared to DF [13] by employing extended causal convolution but lacks direct support for incremental learning. To address this, we modified both DF and Var-CNN by incorporating a k -NN classifier, enabling them to classify an expanded set of websites through retraining. These modified models are referred to as DF* and Var-CNN*, respectively. Although TF, DF*, Var-CNN*, and NetCLR can adapt to classification tasks on new website sets, they lack support for direct incremental learning. Each requires fine-tuning with data from both new and stored old classes, making them sensitive to the distribution similarity between the pre-training and test datasets. To evaluate these methods under different data distributions, we set up experiments comparing TF-Syn vs. DF-Syn and TF-Div vs. DF-Div. For Var-CNN* and NetCLR, results are presented based on pre-training and fine-tuning within the same data distribution.

We conduct CIF evaluations under four settings: dynamic closed-world, dynamic open-world, few-shot Class Incremental Learning, and adversarial defense measures.

A. Dataset

The Wang dataset [11], AWF dataset [12], and DF dataset [13], which are publicly available, are used for experimental evaluation of CIF attacks. The following provides an overview of these datasets:

Wang dataset. Referred to as Wang100, this dataset is composed of a group of websites chosen from a list of sites that are blocked in several countries. Each website within it has 90 examples.

AWF dataset. Built in 2016, the AWF dataset comprises both monitored and unmonitored websites, all of which are picked from the top-ranked sites in the Alexa rankings. The following subsets are involved in our experiments:

- **AWF 100:** Comprises the top 100 monitored websites in the Alexa rankings, and each of these websites supplies 2500 data instances.
- **AWF 200:** Contains an additional 200 monitored websites, with each having 2500 instances.

TABLE I
HYPERPARAMETER SELECTION FOR CIF MODEL

Hyperparameters	Search Range	Final
Learning Rate	[0.0001, ..., 0.002]	0.0005
Weight Decay	[0.0001, ..., 0.01]	0.001
Batch Size	[120, ..., 240]	200
Maximum Length	[3,000, ..., 8,000]	5,000
Time Slot (ms)	[22, ..., 352]	44
Training Epochs	[20, ..., 200]	100
Temperature	[0, ..., 20], step=2	2
Distillation Lambda	[0.1, ..., 2.0], step=0.1	1

TABLE II
THE IMPACT OF TAM TIME SLOT (MS) ON CIF ACCURACY (%)

Time Slot	Undefended	WTF-PAD	FRONT	Walkie-Talkie
22	97.85	92.72	94.23	82.15
33	98.42	93.56	96.85	85.47
44	99.28	92.67	96.19	88.78
88	98.14	89.72	92.13	88.50
176	98.02	88.30	90.57	85.13
352	98.51	87.62	89.35	80.62

- **AWF 775:** Includes 775 more monitored websites, and each provides 2500 instances.
- **AWF 400K:** An open-world dataset containing 9000 unmonitored websites selected from the Alexa rankings, with one example per website.

DF dataset. The DF dataset consists of websites selected from the top Alexa rankings. It includes both undefended and defense-protected traffic traces for 95 monitored websites. Each website provides 1000 examples.

B. Hyperparameter of CIF

To select our model's hyperparameters, we searched within the hyperparameter space, tuning parameters for both the dynamic closed-world and adversarial defense scenarios, using average accuracy as the performance metric. The search ranges and final values of the primary hyperparameters are summarized in Table I. We also investigate key CIF hyperparameters, including the maximum length of traffic traces, the time slot of TAM, and the number of training epochs.

Time Slot of TAM. Table II shows the impact of time slot duration on accuracy in the dynamic closed-world and under adversarial defenses such as WTF-PAD, Front, and Walkie-Talkie. The results indicate that accuracy initially increases with larger time slots, then declines. A small time slot makes the TAM too sparse for effective feature extraction, while a large time slot reduces the TAM's ability to represent the website loading process in detail. Balancing space efficiency and accuracy, we set the time slot to 44 ms.

Maximum Length. Longer website traffic traces generally contain more information, enhancing WF attack performance. As shown in Table III, accuracy improves with increased maximum length, although the growth plateaus beyond

TABLE III
THE IMPACT OF MAX TRAFFIC TRACE LENGTH ON CIF ACCURACY (%)

Max Length	Undefended	WTF-PAD	FRONT	Walkie-Talkie
3000	96.83	90.12	95.05	86.23
4000	98.15	91.34	95.67	87.12
5000	99.31	92.58	96.22	88.76
6000	98.98	92.91	96.35	89.21
7000	99.12	93.15	96.54	89.35
8000	98.87	93.42	96.67	89.02

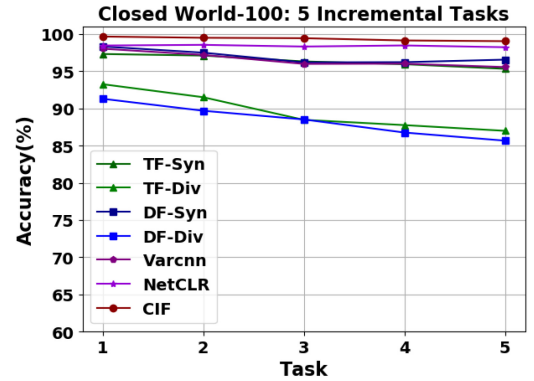


Fig. 4. Accuracy Changes in 5-Step CIL for dynamic Closed-World 100 Websites.

5,000. Additionally, previous studies on website fingerprinting [12], [21], [36] commonly set the maximum length to 5,000. Therefore, we adopt a maximum length of 5,000.

Training Epochs. We set the training epochs to 100 and employed early stopping to prevent overfitting. The validation loss is monitored, with the initial best loss set to infinity and a patience value of 5. If the current loss is lower than the best loss, we update the best loss and reset the counter; otherwise, we increment it. Early stopping is triggered when the counter reaches 5, terminating the training.

C. Dynamic Closed-World Scenario

We first evaluate CIF attacks in a dynamic closed-world scenario, where attackers can dynamically add new websites of interest to the set of monitored websites. Unlike the traditional closed-world setting, where a fixed training dataset of monitored websites is used to train the model, the number of monitored websites in the dynamic closed-world gradually increases. This requires the gradual input of training data from different website classes into the model for training. We tested the CIL scheme in three scenarios, with the total number of dynamically monitored sites set to 100, 200, and 500, respectively.

Performance Metric. The experiments for the CIL scheme [28], [29], [87] evaluate the model's performance based on changes in accuracy over the incremental learning steps, including the First, Last, and average accuracy per step. These metrics are used consistently as evaluation criteria across all scenarios in our experiments.

Experimental Setup. We conducted incremental learning experiments using the AWF100, AWF200, and AWF775 datasets. For each dataset, we trained all classes from scratch

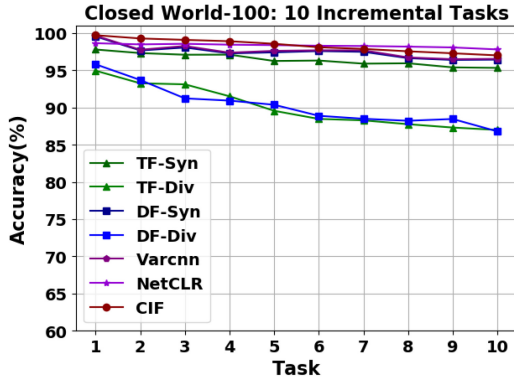


Fig. 5. Accuracy Changes in 10-Step CIL for dynamic Closed-World 100 Websites.

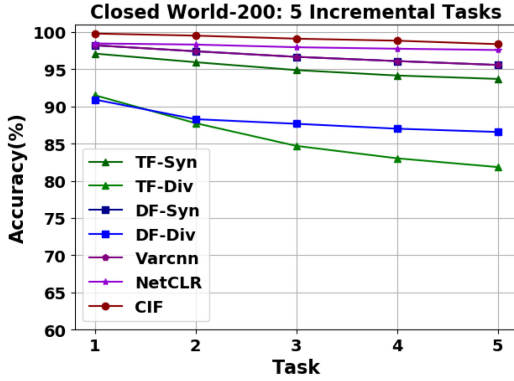


Fig. 6. Accuracy Changes in 5-Step CIL for dynamic Closed-World 200 Websites.

with task splits in two incremental learning steps of 5 and 10. Samples from each website were partitioned into training, validation, and test data at 80%, 10%, and 10%, respectively.

For both TF and NetCLR, the AWF775 dataset was used for model pre-training, followed by fine-tuning with new website traffic traces from the current task. DF* and Var-CNN* follow a similar approach to TF, where the k -NN classifier is retrained with new traffic traces to adapt to the updated set of monitored websites. Specifically, for TF and DF, we evaluated two scenarios: (1) dynamic monitoring datasets with distributions similar to AWF775 for retraining, labeled as TF-Syn and DF-Syn; and (2) datasets with differing distributions, labeled as TF-Div and DF-Div.

Results. Figs. 3–4, 5–6, and 7–8 present accuracy variation curves for incremental steps of 5 and 10 in tasks with 100, 200, and 500 monitored websites, respectively. Tables IV, V, and VI summarize accuracy for First, Last, and Average steps in CIL. Results demonstrate that the CIF attack is highly effective in dynamic closed-world scenarios, achieving an average accuracy exceeding 97.8% across all settings. For 100 monitored websites with 5 incremental tasks, CIF achieves an average accuracy of 99.35%, maintaining 99.04% accuracy in the final step.

In all incremental learning scenarios, CIF consistently outperforms TF, DF*, Var-CNN*, and NetCLR in terms of average accuracy. As the total number of monitored websites increases, CIF's advantage becomes more evident. For

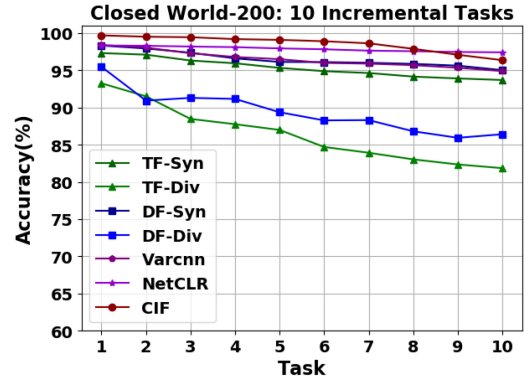


Fig. 7. Accuracy Changes in 10-Step CIL for dynamic Closed-World 200 Websites.

TABLE IV
ACCURACY AT FIRST STEP, LAST STEP, AND AVERAGE ACROSS 5 AND 10 INCREMENTAL TASKS IN DYNAMIC CLOSED-WORLD WITH 100 MONITORED WEBSITES

Acc	5 Tasks			10 Tasks		
	First	Last	Avg	First	Last	Avg
TF-Div	93.26	86.99	89.58	94.95	86.99	90.12
TF-Syn	97.32	95.33	96.40	97.80	95.33	96.45
DF-Div	91.30	85.67	88.38	95.81	86.41	90.01
DF-Syn	98.33	96.56	96.96	99.56	96.44	97.46
Var-CNN	98.73	96.36	97.41	99.66	96.54	97.70
NetCLR	98.46	98.23	98.41	98.65	97.81	98.32
CIF	99.66	99.04	99.35	99.72	97.01	98.34

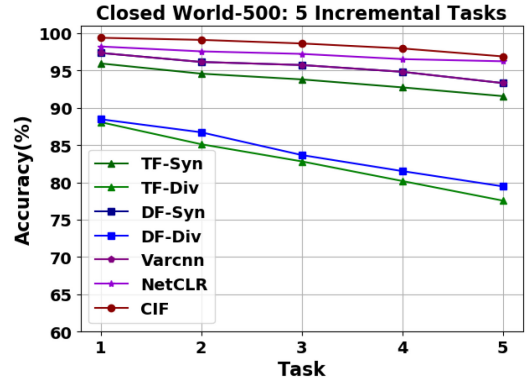


Fig. 8. Accuracy Changes in 5-Step CIL for dynamic Closed-World 500 Websites.

500 monitored websites, CIF's average accuracy surpasses that of TF, DF*, and Var-CNN* by over 3% and maintains this lead in the final incremental phase. In scenarios with five incremental tasks, CIF consistently outperforms NetCLR. For ten tasks, CIF's final accuracy is slightly lower than NetCLR, but the gap diminishes as the number of monitored websites grows.

TF exhibits the lowest performance due to its reliance on complex pairing and ternary constructions, limiting its ability to utilize large training datasets effectively. TF-Syn and DF-Syn perform significantly better than TF-Div and DF-Div, highlighting the impact of distributional differences between pre-training and target task datasets. Additionally, TF, DF*,

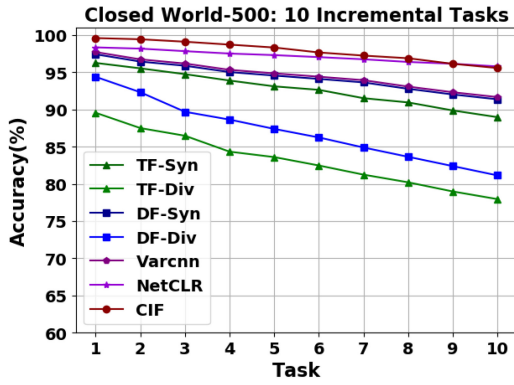


Fig. 9. Accuracy Changes in 10-Step CIL for dynamic Closed-World 500 Websites.

TABLE V
ACCURACY AT FIRST STEP, LAST STEP, AND AVERAGE ACROSS 5 AND 10 INCREMENTAL TASKS IN DYNAMIC CLOSED-WORLD WITH 200 MONITORED WEBSITES

Acc	5 Tasks			10 Tasks		
	First	Last	Avg	First	Last	Avg
TF-Div	91.51	81.85	85.77	93.26	81.85	86.06
TF-Syn	97.11	93.7	95.16	97.32	93.7	95.33
DF-Div	90.93	86.57	88.09	95.48	86.8	89.85
DF-Syn	98.22	95.58	96.79	98.33	95.03	96.5
Var-CNN	98.32	96.23	97.16	98.47	95.15	96.99
NetCLR	98.47	97.59	97.82	98.35	97.42	97.78
CIF	99.80	98.38	99.13	99.70	96.36	98.58

TABLE VI
ACCURACY AT FIRST STEP, LAST STEP, AND AVERAGE ACROSS 5 AND 10 INCREMENTAL TASKS IN DYNAMIC CLOSED-WORLD WITH 500 MONITORED WEBSITES

Acc	5 Tasks			10 Tasks		
	First	Last	Avg	First	Last	Avg
TF-Div	88.08	77.56	82.15	93.26	81.85	86.06
TF-Syn	95.95	91.57	93.13	96.26	88.97	92.89
DF-Div	88.47	79.47	83.37	94.39	81.14	87.63
DF-Syn	97.36	93.32	95.08	97.42	91.37	94.62
Var-CNN	97.30	93.24	95.83	97.72	91.67	95.19
NetCLR	98.22	96.24	97.16	98.35	95.93	97.17
CIF	99.39	96.89	98.39	99.60	95.57	97.87

Var-CNN*, and NetCLR require all samples of each category for retraining, along with retaining all prior samples, making them less suitable for resource-constrained scenarios.

D. Few-Shot Class Incremental Learning

In real-world WF attacks, attackers may not always be able to fully collect training data from each website. Frequently, the attacker needs to update the attack model promptly but lacks sufficient time to gather enough training data. In such cases, the model must perform effectively with only a few-shot sample.

We tested both CIL scenarios, with and without pre-training, employing the Mixtam data augmentation algorithm

TABLE VII
ACCURACY AT FIRST STEP, LAST STEP, AND AVERAGE ACROSS TASKS IN FEW-SHOT CIL WITH AND WITHOUT PRE-TRAINING AND MIXTAM AUGMENTATION

Acc	5 Tasks			50 Pre + 5 Tasks		
	First	Last	Avg	First	Last	Avg
FS10	10.44	6.24	8.64	99.17	35.54	60.60
AU-FS10	69.77	48.88	56.76	99.31	68.84	82.33
FS20	18.77	16.11	17.21	99.3	39.69	63.56
AU-FS20	86.88	79.00	83.75	99.27	84.80	91.47
FS30	22.88	20.66	21.31	99.19	48.06	71.07
AU-FS30	91.66	83.80	87.60	99.22	87.37	93.51

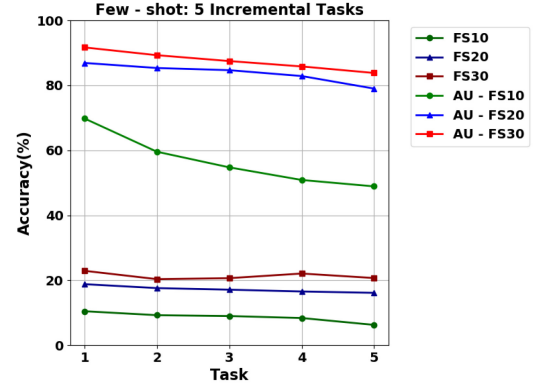


Fig. 10. No Pre-training: Accuracy Changes in 5-Step CIL with Few-shot Samples, with and without Mixtam Augmentation.

to evaluate its effectiveness in few-shot CIL tasks across 100 dynamically monitored websites.

Experimental Setup. The AWF100 dataset was used as the experimental data. In addition to conducting the same 5-step incremental learning experiments as in the dynamic closed-world scenario, we also designed a pre-training scenario for limited samples. In the pre-training scenario, the model was first trained on 50 classes, with the remaining 50 classes divided into 5 and 10-step incremental learning tasks. In real-world environments, attackers may have abundant training data for the initial set of monitored websites but lack sufficient time to collect data for newly monitored websites. Therefore, the pre-trained model's initial 50 classes were trained using sufficient data, selecting 800 samples per website. For the remaining classes, few-shot samples (10, 20, or 30 per website) were selected and expanded to 1,500 samples using the Mixtam algorithm.

Results. To evaluate Mixtam's impact on CIF performance, we conducted comparative experiments under conditions with and without Mixtam augmentation, denoted by AU-FS and FS, respectively. Figs. 10 and 11 illustrate the accuracy variation over CIL steps for scenarios without and with pre-training. Table VII presents the first, last, and per-step average accuracies under both conditions.

In the no-pre-training scenario, CIF performance without Mixtam resulted in an average accuracy below 10% for 10-shot settings and approximately 20% for 30-shot settings. With Mixtam augmentation, average accuracies increased to 56.76%, 83.75%, and 87.60% for 10, 20, and 30 samples,

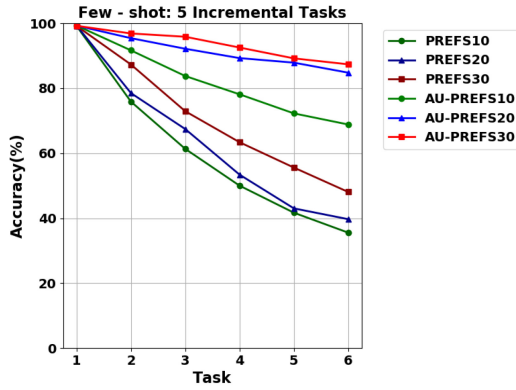


Fig. 11. With Pre-training: Accuracy Changes in 5-Step CIL with Few-shot Samples, with and without Mixtam Augmentation.

respectively, achieving a maximum improvement of over 66%. In the pre-training scenario, CIF accuracies for 20-shot and 30-shot settings were 63.56% and 71.00%, respectively, without Mixtam. When Mixtam was applied, these accuracies increased to 91.47% and 93.51%, representing improvements exceeding 20%.

These results demonstrate that Mixtam data augmentation significantly enhances CIF attack accuracy, especially in cases with limited training samples.

E. CIF Attack Against Defense

In this section, we evaluate the performance of the proposed CIF model on defense-processed traffic traces. For this experiment, we selected several defense methods, including WTF-PAD [35], Walkie-Talkie [50], TrafficSliver [52], FRONT [49], RegulaTor [51], Surakav [53], and Palette [54], to assess the performance of WF attacks on defended traffic. We conducted comparative experiments with Var-CNN* for this evaluation.

The selected defense methods employ various techniques to disrupt traffic patterns. WTF-PAD (WTP) and FRONT insert virtual packets to alter traffic patterns, leveraging natural gaps and Rayleigh distribution timing, respectively, to enhance anonymity. Walkie-Talkie (WT) introduces half-duplex communication to increase obfuscation and strengthen the defense effect. RegulaTor (RegTor) controls packet transmission to maintain high efficiency while limiting resource consumption, creating barriers to traffic tracking. Palette standardizes traffic patterns within the same cluster to prevent attackers from identifying specific websites based on distinct traffic features. Surakav utilizes GANs to diversify transmission patterns and modulate data transmission, effectively resisting WF attacks. TrafficSliver focuses on traffic segmentation, dispersing data flows through network architecture adjustments to reduce the risk of single-point monitoring information leakage.

Experimental Setup. For the closed-world evaluation, we used the DF95 dataset, which contains 95 websites, each with 1,000 undefended traces. The corresponding defense traces were generated using scripts and simulators provided by the defense authors. We tested two scenarios, where the task division of all classes was trained from scratch in two

TABLE VIII
BANDWIDTH AND TIME OVERHEAD OF DEFENSES METHODS (PART 1)

Defenses	WTF-PAD	FRONT	RegulaTor	Walkie-Talkie
Bandwidth	63%	103%	77%	31%
Time	0%	0%	5%	34%

TABLE IX
BANDWIDTH AND TIME OVERHEAD OF DEFENSES METHODS (PART 2)

Defenses	Traffic-Sliver	Surakav	Palette
Bandwidth	0%	80%	84%
Time	0%	6%	9%

TABLE X
ACCURACY CHANGES OF CIF AND VAR-CNN* ACROSS 5 INCREMENTAL TASKS ON DEFENDED TRAFFIC TRACES

Method	Task 1	Task 2	Task 3	Task 4	Task 5
Var-WTP	99.306	98.472	89.167	89.375	82.889
Var-WT	84.861	76.042	69.907	64.375	55.167
Var-BD	96.311	94.117	93.349	90.707	88.406
Var-BWR	31.781	30.953	30.699	29.692	28.825
Var-Front	83.646	77.938	74.790	68.993	64.078
Var-RegTor	48.754	47.618	47.141	45.806	44.631
Var-Surakav	55.080	54.880	54.380	53.430	53.090
Var-Palette	21.310	20.970	20.670	20.360	20.190
CIF-Surakav	81.600	80.910	79.370	78.210	76.920
CIF-Palette	38.570	38.190	37.490	37.250	36.930
CIF-WTP	95.278	94.972	94.759	93.625	92.678
CIF-WT	98.411	97.937	97.734	97.162	96.646
CIF-BWR	81.012	79.817	79.176	77.806	76.564
CIF-BD	97.904	96.740	96.244	94.853	93.609
CIF-Front	95.691	93.848	91.646	91.573	90.637
CIF-RegTor	68.936	68.069	67.719	66.677	65.749

incremental learning steps (5 and 10 steps, respectively). For TrafficSliver [52], we selected the two best network layer splitting strategies: By Direction (BD) and Batched Weighted Random (BWR). The bandwidth and time overhead settings for each defense are shown in Table VIII and Table IX.

Results. Tables X and XI illustrate the accuracy variation curves for the two incremental steps (5 and 10), while Table XII shows the accuracy for the first, last, and average steps in each configuration. The results are based on experiments using Var-CNN* (denoted as Var-) and CIF (denoted as CIF-) for each configuration.

The results indicate that CIF maintains strong performance even under multiple defenses such as WTF-PAD (WTP), Walkie-Talkie (WT), Splitting-Traffic-BD (BD), and Front. For both incremental learning steps, the average accuracy exceeds 90%. Compared to Var-CNN*, our CIF method demonstrates significant advantages. CIF leverages TAM to effectively capture traffic features, enabling it to count the number of incoming and outgoing packets at finer time intervals. This preserves key characteristics and outperforms traditional sequence- or burst-based defenses (e.g., WTF-PAD and Walkie-Talkie).

The Front strategy introduces more bandwidth overhead on the Tor connection, significantly reducing Var-CNN* accuracy, but CIF maintains high accuracy. Traffic-Sliver-BD utilizes

TABLE XI
ACCURACY CHANGES OF CIF AND VAR-CNN* ACROSS 10 INCREMENTAL TASKS ON DEFENDED TRAFFIC TRACE

Method	Task 1	Task 2	Task 3	Task 4	Task 5	Task 6	Task 7	Task 8	Task 9	Task 10
Var-BD	96.406	95.377	93.629	92.187	90.931	90.227	89.528	88.951	87.427	86.237
Var-SBWR	31.426	30.988	30.158	29.499	28.943	28.682	28.424	28.230	27.533	27.015
Var-WTP	99.444	93.333	87.278	91.750	89.444	90.185	92.024	84.931	84.414	87.694
Var-Front	88.945	86.739	83.056	80.107	77.603	76.229	74.883	73.786	70.916	68.737
Var-WT	87.222	83.889	82.407	77.431	76.667	71.111	67.817	63.160	58.426	55.194
Var-RegTor	48.579	47.925	46.842	45.948	45.169	44.721	44.277	43.907	42.979	42.253
Var-Surakav	54.810	54.120	54.670	54.150	53.640	53.230	52.930	52.020	51.470	51.260
Var-Palette	21.310	21.030	20.840	20.630	20.460	20.330	19.980	19.900	19.510	19.430
CIF-BD	98.609	97.998	96.898	95.996	95.214	94.807	94.402	94.079	93.093	92.334
CIF-BWR	83.890	83.135	81.974	80.985	80.101	79.533	78.969	78.476	77.440	76.595
CIF-WTP	96.667	95.800	95.330	94.917	92.630	92.110	91.667	92.090	91.895	91.361
CIF-Front	97.768	96.913	95.476	94.281	93.235	92.635	92.040	91.543	90.271	89.268
CIF-WT	97.556	97.056	97.185	95.611	95.022	94.352	94.238	94.708	94.543	94.422
CIF-RegTor	70.045	69.415	68.446	67.620	66.882	66.407	65.937	65.525	64.660	63.954
CIF-Surakav	81.200	80.160	79.710	79.470	78.330	76.810	75.970	74.460	73.880	72.940
CIF-Palette	38.570	37.820	37.480	36.980	36.280	35.850	35.590	35.070	34.720	34.350

separate circuits for packet reception and transmission, allowing attackers to capture packets in only one direction. Both Var-CNN* and CIF can leverage packet timing characteristics, resulting in minimal accuracy degradation. BWR employs a weighting strategy to select protection nodes for Tor messages, increasing the time intervals between message batches and reducing inter-message correlation. In contrast, Surakav uses GANs to generate diversified transmission patterns, combining burst adjustments and random responses to effectively minimize the leakage of timing and packet features. Under both defenses, CIF still achieves an average accuracy of nearly 80%, significantly higher than Var-CNN*'s 30%.

Palette introduces traffic feature obfuscation and randomness, making it difficult for both Var-CNN* and CIF to extract stable features, leading to a significant drop in accuracy for both models.

F. Dynamic Open-World Scenario

In previous experiments, we investigated the impact of CIF attacks on performance in multiple scenarios within the dynamic closed-world. However, real-world network environments often involve users visiting websites beyond the monitored ones. To address this, we evaluated the CIF attack in a more realistic dynamic open-world setting. This evaluation involves classifying whether a traffic trace belongs to the set of monitored websites and, if so, further identifying the specific website within the monitored set.

Experimental Setup. We conduct experimental tests in a fairly large open-world environment, a scenario in which users have access to 100,000 non-monitored sites in addition to the websites in the monitored sites. We take 99 monitored websites in the AWF100 dataset as the 99 classes in the open-world multi-classification, with 1,500 samples per website. We take the 100,000 non-monitored websites in the AWF400K dataset as a class of non-monitored websites in the Open-World multi-classification, with one sample per website. For these 100 websites, we consider CIL in two steps (5 and 10). For this scenario, we set the number of samples in the test set for the

TABLE XII
FIRST STEP, LAST STEP, AND AVERAGE ACCURACY ACROSS 5 AND 10 INCREMENTAL TASKS AGAINST TRAFFIC DEFENSES

Acc	5 Tasks			10 Tasks		
	First	Last	Avg	First	Last	Avg
Var-WTP	99.30	82.89	91.84	99.44	87.69	90.05
Var-WT	84.86	55.16	70.07	87.22	55.19	72.33
Var-BD	96.31	88.40	92.57	96.40	86.37	91.09
Var-BWR	31.78	28.82	30.39	31.42	27.01	29.09
Var-Front	83.64	64.07	73.88	88.94	68.73	78.10
Var-RegTor	48.75	44.63	46.79	48.57	42.25	45.26
Var-Surakav	55.08	53.09	54.17	54.81	51.26	53.23
Var-Palette	21.31	20.19	20.70	21.32	19.43	20.34
CIF-WTP	95.27	92.67	94.26	96.66	91.36	93.45
CIF-WT	98.41	96.64	97.57	97.55	94.42	95.46
CIF-BD	97.90	93.60	95.87	98.60	92.33	95.34
CIF-BWR	81.01	76.56	78.87	83.89	76.59	80.11
CIF-Front	95.69	90.63	92.67	97.76	89.26	93.34
CIF-RegTor	68.93	65.74	67.43	70.04	63.95	66.88
CIF-Surakav	81.6	76.92	79.40	81.20	72.94	77.29
CIF-Palette	38.57	36.93	37.69	38.57	34.35	36.27

TABLE XIII
ACCURACY AT FIRST STEP, LAST STEP, AND AVERAGE ACROSS 5 AND 10 INCREMENTAL TASKS IN DYNAMIC OPEN-WORLD

Acc	5 Tasks			10 Tasks		
	First	Last	Avg	First	Last	Avg
TF	76.95	56.14	66.09	91.14	56.14	70.63
DF-Div	89.89	81.36	84.65	91.73	81.43	84.92
DF-Syn	96.15	91.73	93.62	97.76	91.39	93.75
Var-cnn	99.44	97.88	98.31	97.18	91.71	94.73
NetCLR	84.65	79.71	80.54	88.71	79.71	80.98
CIF	99.44	97.88	98.31	98.7	95.66	96.74

class of non-monitored websites to be the same as the total number of samples in the test set for all the remaining classes.

In this scenario, we compared CIF with TF, DF*, Var-CNN*, and NetCLR [39]. Due to minimal performance differences in TF across datasets with similar distributions, only one result for TF is presented.

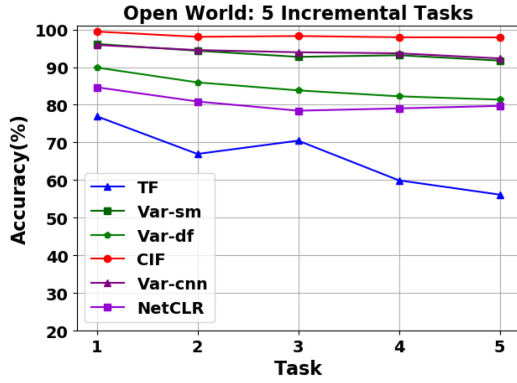


Fig. 12. Accuracy Changes in 5-Step CIL for dynamic Open-World.

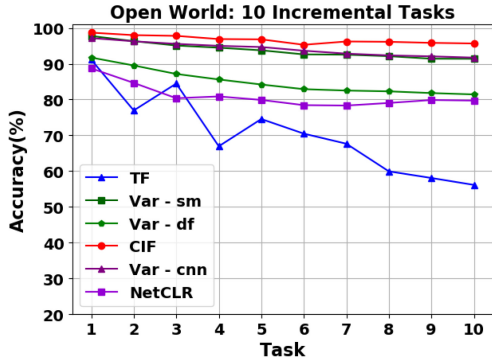


Fig. 13. Accuracy Changes in 10-Step CIL for dynamic Open-World.

Results. Figs. 12 and 13 depict the accuracy curves for 5 and 10 incremental steps, respectively. Table XIII summarizes the accuracy in the First, Last, and Average across steps.

The results demonstrate that the CIF attack remains effective in open-world scenarios, achieving an average accuracy of over 96.7%. For the 5-step incremental setup, it achieves an average accuracy of 98.31% and maintains 97.88% accuracy in the last step, exceeding the performance of other methods by more than 6%. In the 10-step setup, CIF maintains a performance advantage of over 4% in the last step compared to the other methods.

Notably, the CIF attack exhibits a more pronounced performance advantage in the dynamic open-world compared to the dynamic closed-world. In the open-world, TF, DF*, and Var-CNN* are more susceptible to factors such as uneven data distribution, feature space variation, and inter-class similarity. In contrast, CIF's dynamic tuning and memorization mechanisms are better equipped to handle these challenges. TF and NetCLR, in particular, perform poorly in this scenario. TF struggles with effectively utilizing the extensive training data from non-monitored classes, leading to severe class imbalance and reduced accuracy. Similarly, NetCLR's reliance on NETAUGMENT and contrastive learning makes it vulnerable to disruptions from the diverse traffic data in the non-monitored category, degrading the quality of its initial learning and subsequent fine-tuning, resulting in significant accuracy drops.

VI. DISCUSSION

In this section, we explore the proposed CIF attack and suggest potential directions for future work.

The CIF model, leveraging a dynamic scaling architecture, is capable of continuously acquiring new information from emerging website classes without the need for complete retraining, thereby exhibiting remarkable scalability. This innovative CIF attack approach addresses the limitations of prior WF attack methods, which lacked continuous learning capabilities in open environments. As a result, CIF effectively satisfies the requirements of attackers for expanding surveillance scopes and updating models in dynamic settings.

While our designs, including DF*, Var-CNN*, and TF, can be adapted for website expansion tasks through classifier retraining, their performance is heavily constrained by the similarity of dataset distributions. Significant discrepancies between pre-trained data and new task data often lead to pronounced declines in accuracy. Furthermore, TF and NetCLR exhibit severe performance degradation under highly dynamic open-world scenarios. Unlike CIF, these methods rely on the continuous storage of all samples across classes and necessitate the reprocessing of the entire dataset during each retraining cycle, making them impractical for resource-constrained environments.

CIF has consistently demonstrated outstanding performance across a variety of experimental scenarios. Nonetheless, further research is essential to address the complexities inherent in more challenging real-world applications.

Higher Few-shot Performance. Facing the rapidly changing network environment, attackers must efficiently collect the latest network traffic traces and promptly update their attack models. This requirement highlights the necessity for CIF attacks to possess robust few-shot learning capabilities. In real scenarios, we may only be able to collect limited data for some websites in a short period (e.g., less than 10 training samples per class). Therefore, in addition to the data enhancement methods already designed, we need to explore how we can better adapt to the few-shot scenario in the CIF framework species.

Fine-Grained website fingerprinting. Current research on WF attacks often focuses exclusively on the homepage traffic of a website as a representative identifier for the entire site. This simplification overlooks the diversity of subpages, which is inconsistent with the complexity of real-world Web browsing environments. Subpages of the same website typically share similar layout structures, follow consistent encryption protocols, and use comparable parameter configurations. These similarities result in high convergence in the characteristics and client-server interaction patterns of the Web traffic they generate.

Exploring fine-grained Web fingerprinting techniques is essential for accurately modeling real-world scenarios and uncovering more profound insights into user privacy. Our research will focus on advancing the development of fine-grained CIF attack techniques to better align with the practical demands of real-world applications.

VII. CONCLUSION

In this study, we investigate implementing WF attack methods with continuous learning capability through incremental class learning. This approach addresses the limitations of existing attack models, which lack continuous learning for more challenging real-world application scenarios. We propose CIF attacks based on a dynamically scaling architecture that continuously learns knowledge from new classes without catastrophic forgetting. This design enables the attack model to scale and adapt to dynamically open real-world environments. To gain insight into the performance of CIF attacks in reality, we have recently set up dynamic closed-world and open-world scenarios. The experimental results indicate that the CIF attack achieves high accuracy during the CIL process, maintaining an average accuracy above 97.8% in the dynamic closed-world and remaining effective in the dynamic open-world with an average accuracy exceeding 96.7%. In addition, we also demonstrate that the CIF attack achieves good performance when faced with few-shot scenarios. According to these findings, attackers can utilize new data update models to cope with dynamically open real-world environments with limited resources.

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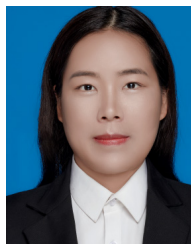
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